INVERTIS UNIVERSITY BAREILLY

PROJECT REPORT

BANK MARKETING CAMPAIGN ANALYSIS

Descriptive analytics, statistical validation and an interactive dashboard

Project ID DA-014 | Marketing Analytics

Project detail	Information
University	Invertis University, Bareilly
Programme	BCA AI
Semester / Section	I / C
Group ID	BCA_AI_14
Mentor	Mr. Ankit
Academic year	2026-27

Department and Course Details

This project was prepared by Group BCA_AI_14 of the BCA AI programme (Semester I, Section C) at Invertis University, Bareilly, under the guidance of Mr. Ankit.

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Acknowledgement

We express our sincere gratitude to our mentor, Mr. Ankit, for his guidance, feedback and encouragement throughout Project DA-014. We also thank Invertis University, Bareilly, and the BCA AI faculty for providing the learning environment and resources needed to complete this work.

We acknowledge the creators and maintainers of the UCI Machine Learning Repository and the authors of the Bank Marketing dataset, S. Moro, P. Cortez and P. Rita, whose public data made this analysis possible.

Abstract

Bank telemarketing teams must decide whom to contact, through which channel and how often, while balancing campaign cost and customer experience. This project analyses the public UCI Bank Marketing dataset (bank-additional-full.csv: 41,188 telephone contact records of a Portuguese bank, May 2008 to November 2010) to describe how responses to a term-deposit campaign vary by communication channel, contact month, weekday, contact frequency, previous campaign outcome, job and age.

A shared Python analysis module drives both a Jupyter notebook and an interactive Streamlit dashboard, so every count and rate is recalculated consistently for any filtered cohort. Of 41,188 records, 4,640 subscribed, an overall response rate of 11.27% (95% Wilson interval 10.96% to 11.57%). Cellular contacts responded at 14.74% against 5.23% for telephone; records with a previous successful campaign responded at 65.11%; and the response rate fell from 13.04% for one contact to 6.14% for five or more.

Confidence intervals, chi-square tests and Cramér's V, computed from the aggregated counts, show that these differences are far larger than sampling noise would explain, while effect sizes separate meaningful gaps (previous outcome, month, channel) from negligible ones (weekday). All findings are unadjusted historical associations; no causal claim is made and no predictive model is built.

Keywords: bank marketing, term deposit, response rate, descriptive analytics, Wilson interval, chi-square test, Streamlit dashboard.

1. Introduction

1.1 Background

Direct marketing by telephone remains an important way for banks to promote savings products such as term deposits, a fixed-period deposit paying an agreed interest rate. Each call consumes staff time and may affect how customers feel about the bank, so campaign managers need to understand where outreach produces responses and where effort is spent with little return.

This project examines a public, record-level dataset of telephone contacts made by a Portuguese bank. It summarises observed responses across communication channels, contact periods, previous campaign outcomes, contact frequency and customer segments, and supports the summary with confidence intervals and significance tests.

1.2 Why this project matters

Percentages quoted without their denominators are easy to misread. A 50% rate based on 182 records is a very different kind of evidence from a 6% rate based on 13,769 records. The project therefore pairs every rate with its record count and an uncertainty interval, and provides an interactive dashboard where any cohort can be inspected and recalculated on demand.

1.3 Scope and interpretation

- The target **y** records whether a term deposit was subscribed (`yes`) or not (`no`).
- Each row is treated as a **contact record**. The supplied file has no unique customer identifier, so record counts are not verified counts of unique people.
- Rates describe **historical associations**. They do not establish that channel, month or contact count caused an outcome.
- The work is **descriptive**. No prediction model is trained, validated or deployed.

2. Problem Statement and Objectives

2.1 Problem statement

Campaign comparisons can be misleading when percentages are presented without their denominators, and when apparent differences are not tested against random variation. This project provides an inspectable, filterable summary that pairs response rates with record counts and uncertainty intervals, and makes the selected cohort visible at every step.

2.2 Objectives

1. Document and inspect the public dataset and its provenance.
2. Define the target and the response-rate denominator consistently across the notebook and the dashboard.
3. Compare outcomes by channel, contact period, previous outcome, campaign contact count and customer segment.
4. Quantify uncertainty with Wilson confidence intervals and test group differences with chi-square tests and effect sizes.
5. Provide interactive filters, charts, a data preview and calculated findings in a Streamlit dashboard.
6. Keep the notebook and dashboard calculations aligned through one shared analysis module.
7. Avoid causal or predictive claims that the descriptive implementation cannot support.

2.3 Guiding questions

Table 2. *Questions addressed in this report*

No.	Question	Section
Q1	What proportion of contact records ended in a subscription?	8
Q2	Does the response rate differ between cellular and telephone contacts?	8
Q3	How does the rate change with the number of contacts in the campaign?	8
Q4	How do the contact month, weekday and previous campaign outcome relate to response?	9
Q5	Which job and age groups respond more or less often?	10
Q6	Are the observed differences larger than random variation, and how large are they?	11

3. Literature Review

3.1 Source and context

The UCI Machine Learning Repository describes Bank Marketing as data from direct marketing campaigns of a Portuguese banking institution. The campaigns were based on phone calls, and the full “additional” file spans May 2008 to November 2010 and records term-deposit subscription as the target y [3].

3.2 Related work

Moro, Laureano and Cortez (2011) applied the CRISP-DM methodology to bank direct marketing data, illustrating a structured data-mining process from business understanding to evaluation [1]. Moro, Cortez and Rita (2014) later studied a data-driven approach to predicting the success of bank telemarketing, including temporal evaluation of models and the selection of relevant features [2]. These papers are frequently used for classification research.

Their predictive models and performance figures are **context only** and are not results of this project. The project files contain no trained estimator, train/test split or predictive score.

3.3 A caution on call duration

The dataset documentation warns that the call duration is known only after a call ends and strongly influences the outcome, so it should be discarded in any realistic pre-call predictive model [3]. This project does not predict, so duration is retained only as a descriptive field; the caution is carried forward to the future-scope section.

3.4 Statistical background

A response rate estimated from a sample is uncertain. The Wilson score interval is a recommended way to express that uncertainty for proportions, particularly for small groups or rates far from 50% [4][5]. The chi-square test of independence checks whether the outcome distribution differs across categories, and Cramér’s V [6] expresses the size of the association on a 0 to 1 scale, which matters because very large samples can make even trivial differences “significant”.

3.5 Approach in this project

The implementation emphasises transparent group counts, response rates, interval estimates, filters and limitations rather than predictive modelling or causal inference.

4. Dataset Description

The project uses UCI's **bank-additional-full.csv** variant. The source CSV is semicolon-delimited and contains 41,188 records and 21 columns: 20 inputs and the target y. The records concern a Portuguese bank and span May 2008 through November 2010. The supplied schema has no customer ID.

Table 3. *Field groups in the dataset*

Field group	Examples / meaning
Target	y: yes = term deposit subscribed; no = not subscribed
Demographic	age, job, marital status, education
Financial / household	default, housing loan, personal loan
Current contact	contact channel, month, weekday, duration, campaign contacts
Previous campaign	pdays, previous contacts, poutcome
Economic context	employment variation rate, consumer price and confidence indices, Euribor 3-month rate, number employed

Table 4. *Dataset properties*

Property	Observed value
Variant	bank-additional-full.csv
Records retained	41,188
Columns	21 (20 inputs + target y)
yes / no	4,640 / 36,548
Class balance	11.27% yes, 88.73% no (imbalanced)
Context	Portugal; May 2008 to November 2010
Unique customer ID	Not present

"Unknown" values are retained as explicit categorical labels. UCI reports no missing values in the published full file; the project cleaning converts a literal "?" to missing if present. A full field dictionary is given in Appendix A.

> **Why the imbalance matters.** With only about one record in nine being a "yes", a rule that always predicted "no" would be right 88.73% of the time. This is another reason the project reports group-level rates and counts rather than a single accuracy figure.

5. System Architecture Diagram

A shared analysis module feeds the interactive dashboard and the notebook, keeping target validation and rate calculations aligned. The layers below move from the public source data to the deliverables.

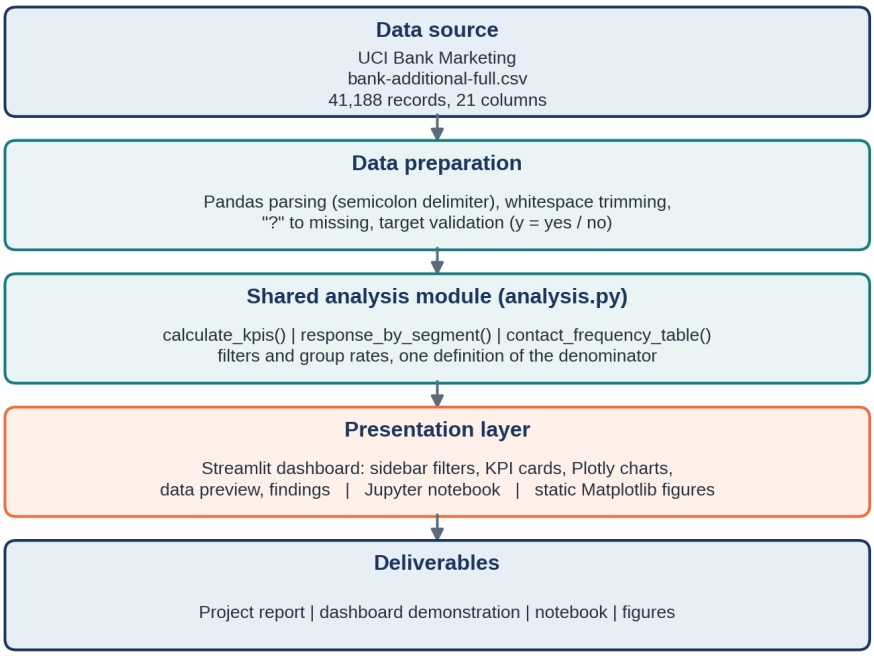


Figure 1. *System architecture of the DA-014 descriptive analytics prototype.*

Table 5. *Function of each layer*

Layer	Function
Public UCI data	bank-additional-full.csv; 41,188 contact records
Data preparation	Pandas parsing, normalisation and target validation
Shared analysis	KPIs, filtering, group rates, contact-frequency summaries
Interactive dashboard	Streamlit filters, Plotly charts, data preview, findings
Deliverables	Notebook, dashboard, figures, report

Design principle. Both the dashboard and the notebook import the same functions, so a rate shown in a dashboard card cannot differ from the one printed in the notebook for the same cohort.

6. Methodology and Project Workflow

The project uses Python, Pandas, Streamlit and Plotly. A shared analysis module reads and normalises the data, validates the target, calculates KPIs, applies filters and groups outcomes. Static Matplotlib charts provide an additional view of the core comparisons. Statistical validation (Section 11) was computed from the aggregated counts using SciPy.

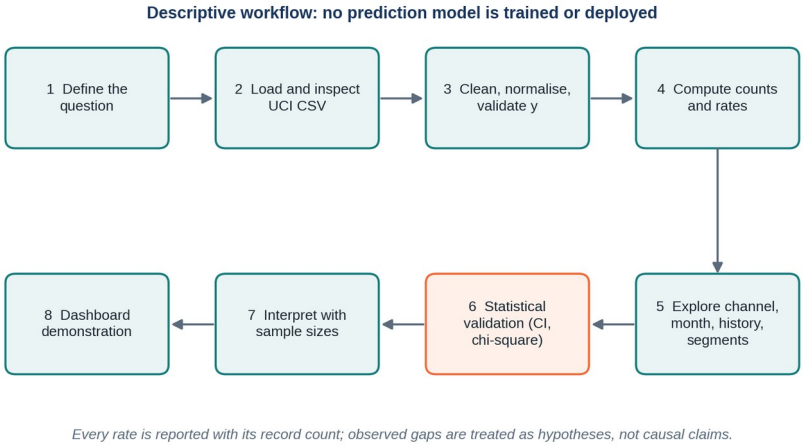


Figure 2. *Data-to-insight workflow. No prediction model is trained or deployed.*

Table 6. *Workflow steps*

Step	Activity
1	Define the question: describe term-deposit campaign response patterns.
2	Load and inspect the semicolon-delimited full UCI dataset.
3	Normalise values and retain rows with valid y = yes / no.
4	Calculate response counts and rates and group by available fields.
5	Explore channel, month, previous outcome, contact frequency and segments.
6	Validate statistically: Wilson intervals, chi-square tests, effect sizes.
7	Interpret with sample sizes; avoid causal claims.
8	Demonstrate by filtering, reviewing charts and clearing filters.

Table 7. *Measure definitions*

Measure	Definition
Successful response	Count of records with y = yes
Overall response rate	yes records ÷ all valid yes/no target records in the cohort
Segment response rate	yes records in group ÷ records in the same group
Mean campaign contacts	Mean of the campaign field; includes the last contact
Filter behaviour	Selected categories and inclusive numeric ranges restrict the cohort before recalculation
95% Wilson interval	$(\hat{p} + z^2/2n \pm z\cdot\sqrt{(\hat{p}(1-\hat{p})/n + z^2/4n^2)}) \div (1 + z^2/n)$, with z = 1.96
Full-data calculation	4,640 ÷ 41,188 = 11.2654%, displayed as 11.27%

7. Data Preprocessing

Table 8. *Preprocessing steps and observed handling*

Step / check	Observed handling
Delimiter	Semicolon; initial spaces skipped
Column names	Whitespace trimmed
Text fields	Whitespace trimmed; literal “?” mapped to missing
Categories	“unknown” normalised to an explicit Unknown label; retained
Target	Normalised to lowercase and validated as yes/no
Invalid target rows	0 excluded
Rows retained	41,188
Missing cells after cleaning	0
Exact duplicate rows	12 observed; retained because no unique key identifies them as erroneous
Unknown category cells	12,718 across 10 categorical fields (cell occurrences, not unique records)

The full valid-target cohort is preserved. Exact duplicates are disclosed rather than silently removed, because an identical row cannot be adjudicated without a customer or contact key: two different customers can legitimately share the same recorded attributes.

8. Exploratory Data Analysis

EDA first checks the target distribution, then compares rates across selected fields. Percentages are interpreted with group sizes and 95% Wilson intervals, particularly where volumes are uneven.

Table 9. *Headline measures*

Measure	Observed value	Definition
Contact records	41,188	Valid yes/no target rows
Successful responses	4,640	y = yes
Non-success records	36,548	y = no
Overall response rate	11.27%	4,640 ÷ 41,188 (95% interval 10.96% to 11.57%)
Mean campaign contacts	2.57	Includes the last contact

8.1 Target distribution

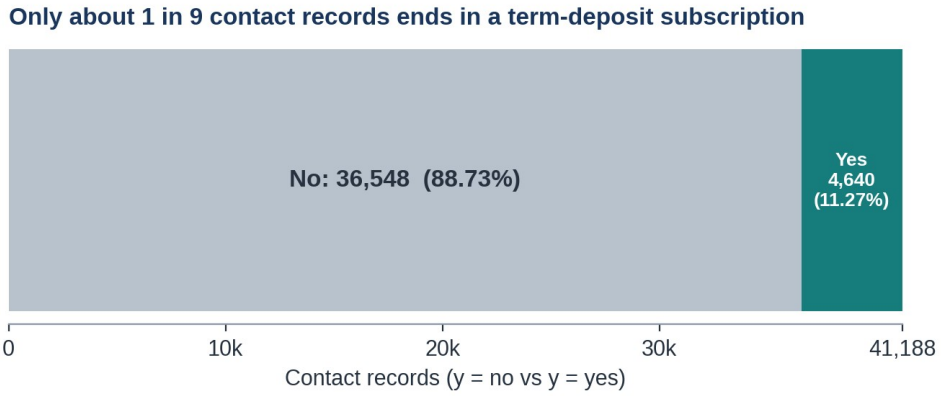


Figure 3. *Overall outcome distribution: 4,640 of 41,188 records are y = yes (11.27%).*

8.2 Communication channel

Table 10. *Response by communication channel*

Channel	Records	Responses	Rate	95% interval (%)
cellular	26,144	3,853	14.74%	14.31 to 15.17
telephone	15,044	787	5.23%	4.89 to 5.60

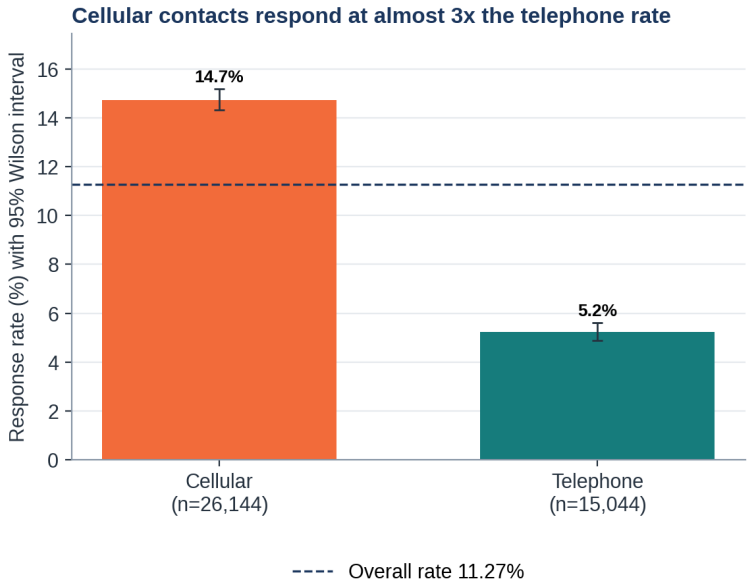


Figure 4. *Channel comparison: cellular 14.74%; telephone 5.23%. Error bars show 95% Wilson intervals.*

The two intervals are far apart and do not overlap. Cellular contacts account for 63.5% of all records but 83.0% of all successful responses. This is a historical, unadjusted association: the channel may be correlated with customer type, month or campaign wave.

8.3 Campaign contact frequency

Table 11. *Response by number of contacts in the campaign*

Contacts	Records	Responses	Rate	95% interval (%)
1	17,642	2,300	13.04%	12.55 to 13.54
2	10,570	1,211	11.46%	10.86 to 12.08
3	5,341	574	10.75%	9.94 to 11.61
4	2,651	249	9.39%	8.34 to 10.56
5+	4,984	306	6.14%	5.51 to 6.84

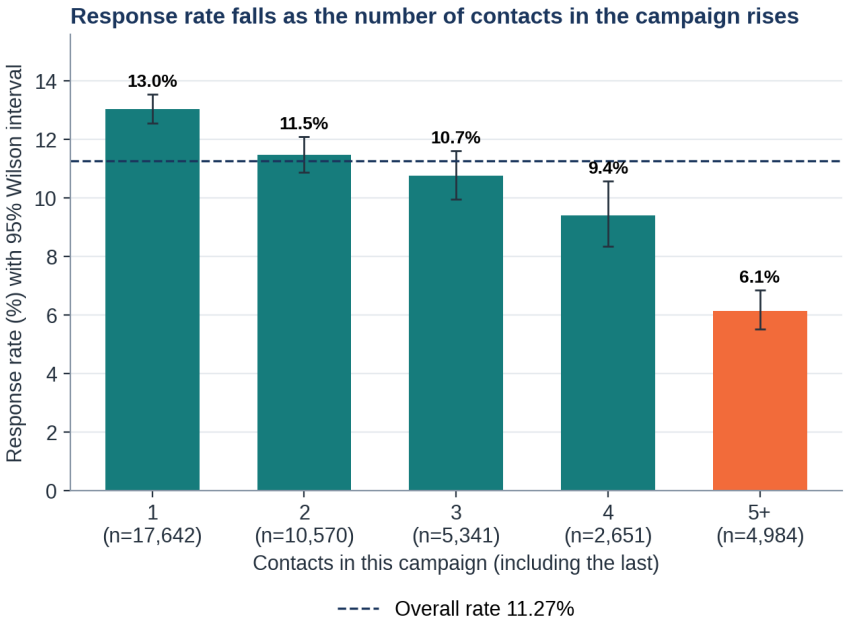


Figure 5. *Response rate by current-campaign contact count, including the last contact.*

The displayed rate declines across the contact-count groups. This descriptive pattern is not evidence that repeated contacts cause lower response; outreach assignment and customer history may differ across groups, for example customers who are persistently hard to reach may simply be called more often.

9. Contact-Period Comparisons

Contact month and weekday comparisons are unadjusted. Group rates should be read together with their denominators and intervals.

9.1 Month

Table 12. *Response by contact month*

Month	Records	Responses	Rate	95% interval (%)
Mar	546	276	50.55%	46.37 to 54.72
Apr	2,632	539	20.48%	18.98 to 22.06
May	13,769	886	6.43%	6.04 to 6.86
Jun	5,318	559	10.51%	9.72 to 11.36
Jul	7,174	649	9.05%	8.40 to 9.73
Aug	6,178	655	10.60%	9.86 to 11.39
Sep	570	256	44.91%	40.88 to 49.02
Oct	718	315	43.87%	40.28 to 47.52
Nov	4,101	416	10.14%	9.26 to 11.11
Dec	182	89	48.90%	41.74 to 56.11

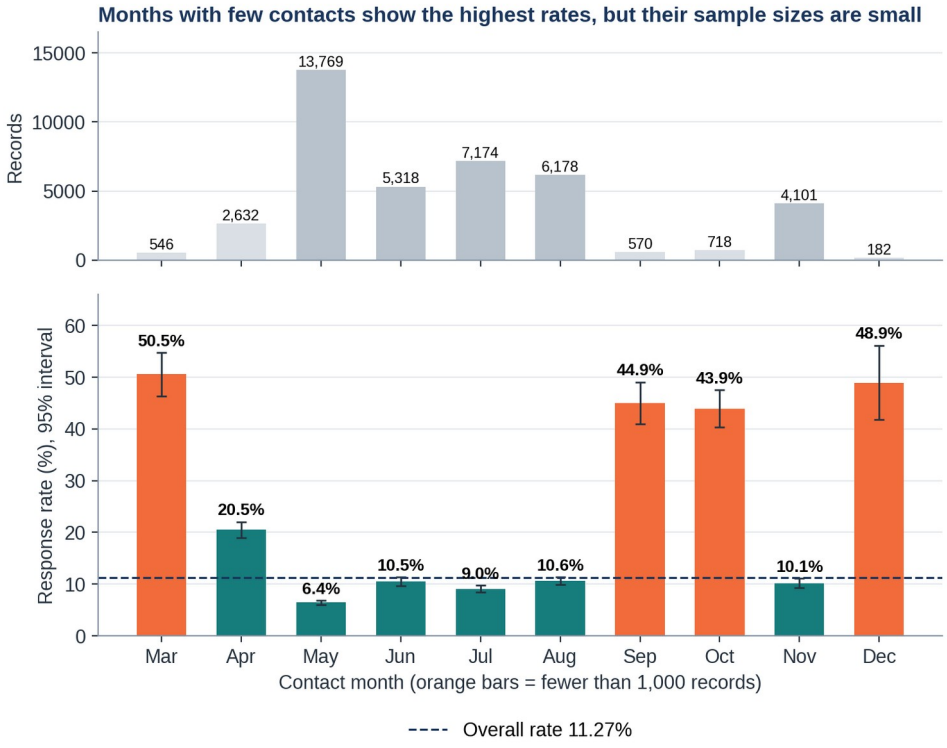


Figure 6. *Response rate by month with monthly record volumes. Orange bars mark months with fewer than 1,000 records.*

Monthly sizes range from 182 records in December to 13,769 in May. March, September, October and December together hold only 2,016 records (4.9% of the file) but 936 successful responses (20.2%), a combined rate of 46.4%. May, in contrast, holds 33.4% of all records and

yields 19.1% of the responses (6.43%). The table does not establish seasonality or explain the differences; the month field may mix campaign waves, targeting choices and the economic conditions of that period.

9.2 Weekday

Table 13. *Response by contact weekday*

Weekday	Records	Responses	Rate	95% interval (%)
Mon	8,514	847	9.95%	9.33 to 10.60
Tue	8,090	953	11.78%	11.10 to 12.50
Wed	8,134	949	11.67%	10.99 to 12.38
Thu	8,623	1,045	12.12%	11.45 to 12.82
Fri	7,827	846	10.81%	10.14 to 11.52

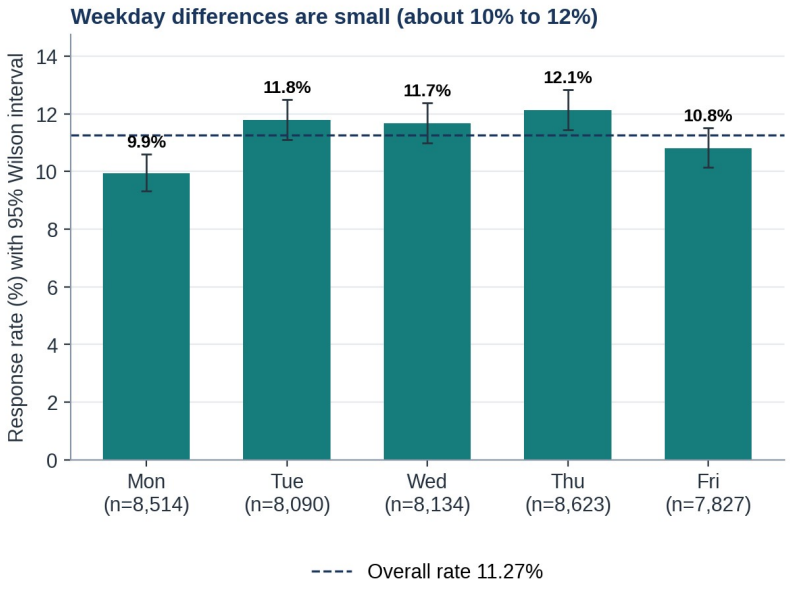


Figure 7. *Response rate by weekday. Differences are small.*

Rates range from 9.95% on Monday to 12.12% on Thursday. Although the chi-square test detects a difference because the sample is large, the effect size is negligible (Cramér's V = 0.025), so weekday is a weak lead for planning.

9.3 Previous campaign outcome

Table 14. *Response by previous campaign outcome*

Outcome	Records	Responses	Rate	95% interval (%)
success	1,373	894	65.11%	62.55 to 67.59
failure	4,252	605	14.23%	13.21 to 15.31
nonexistent	35,563	3,141	8.83%	8.54 to 9.13

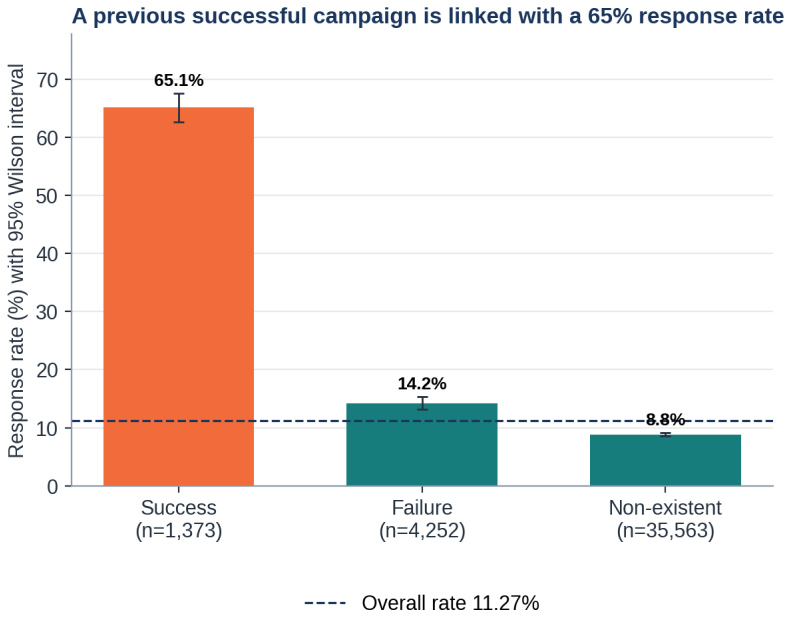


Figure 8. *Response rate by recorded previous campaign outcome.*

Records with a previous campaign success have a 65.11% rate, compared with 14.23% for previous failure and 8.83% for no previous campaign. Previous-success records are only 3.3% of the file yet provide 19.3% of all responses. This is a descriptive association: customers who said yes before may be more receptive for reasons unrelated to the earlier campaign itself.

10. Customer Segment Comparisons

These unadjusted comparisons are descriptive, not targeting rules. Different sample sizes and overlapping characteristics may affect observed rates.

10.1 Job category

Table 15. *Response by job category*

Job	Records	Responses	Rate	95% interval (%)
student	875	275	31.43%	28.44 to 34.58
retired	1,720	434	25.23%	23.24 to 27.34
unemployed	1,014	144	14.20%	12.19 to 16.48
admin.	10,422	1,352	12.97%	12.34 to 13.63
management	2,924	328	11.22%	10.12 to 12.41
Unknown	330	37	11.21%	8.24 to 15.07
technician	6,743	730	10.83%	10.11 to 11.59
self-employed	1,421	149	10.49%	9.00 to 12.19
housemaid	1,060	106	10.00%	8.34 to 11.95
entrepreneur	1,456	124	8.52%	7.19 to 10.06
services	3,969	323	8.14%	7.33 to 9.03
blue-collar	9,254	638	6.89%	6.40 to 7.43

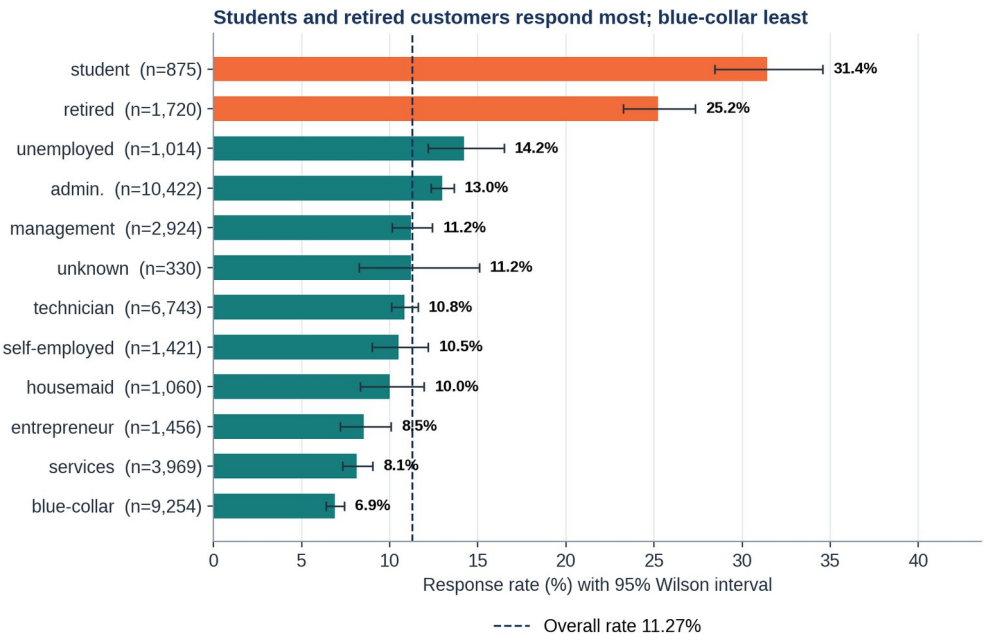


Figure 9. *Response rate by job category (sorted), with 95% Wilson intervals.*

Students (31.43%) and retired customers (25.23%) respond most often; blue-collar (6.89%) and services (8.14%) least. The “Unknown” job group has only 330 records, and its wide interval (8.24% to 15.07%) shows why it should not be ranked against larger groups.

10.2 Age bands

Table 16. *Response by age band*

Age band	Records	Responses	Rate	95% interval (%)
Under 30	5,669	922	16.26%	15.33 to 17.25
30–39	16,938	1,715	10.13%	9.68 to 10.59
40–49	10,526	834	7.92%	7.42 to 8.45
50–59	6,862	697	10.16%	9.46 to 10.89
60+	1,193	472	39.56%	36.83 to 42.37

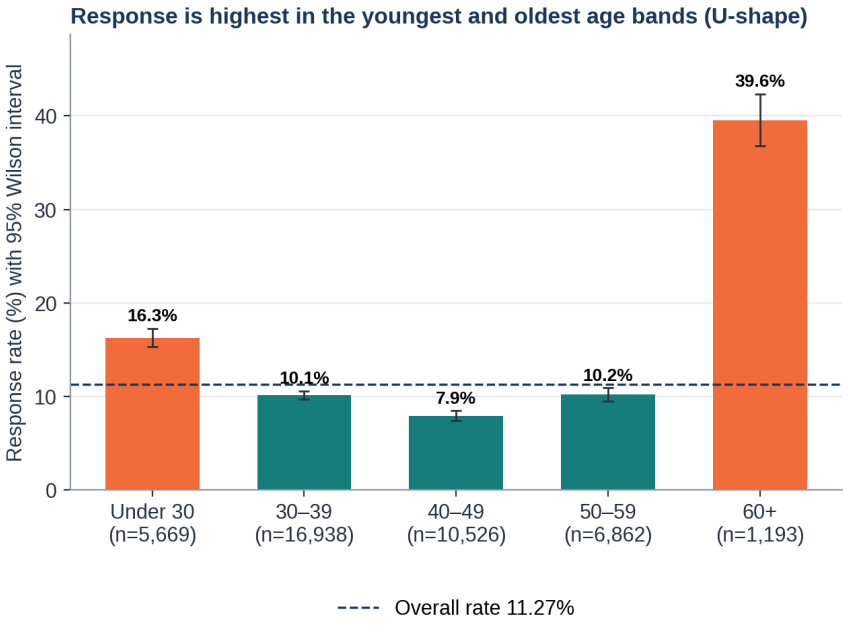


Figure 10. *Response rate by age band: highest under 30 and at 60+.*

The 60+ band contains 1,193 records with a 39.56% rate; Under 30 contains 5,669 records with a 16.26% rate; the middle bands sit between 7.92% and 10.16%. The pattern is U-shaped. It does not show that age alone predicts an individual outcome, and the age effect overlaps with job (students and retired customers sit at the two ends).

11. Statistical Validation

To test whether observed differences are larger than sampling noise, the aggregated counts in Sections 8 to 10 were analysed with standard methods in SciPy. All calculations use only the grouped counts already reported in this document; the script is given in Appendix B.

11.1 Overall rate

The full-data rate of 11.27% has a 95% Wilson interval of 10.96% to 11.57%. With 41,188 records the estimate is tight; the real uncertainty lies in the smaller groups.

11.2 Do the groups differ? Chi-square tests

Table 17. *Chi-square tests of independence between each field and the outcome y*

Field	Groups	χ^2	df	p-value	Cramér's V	Strength
Weekday	5	26.1	4	< 0.001	0.025	Negligible
Contact count	5	197.5	4	< 0.001	0.069	Weak
Channel	2	863.3	1	< 0.001	0.145	Small to moderate
Job	12	961.2	11	< 0.001	0.153	Small to moderate
Age band	5	1,245.5	4	< 0.001	0.174	Small to moderate
Month	10	3,101.1	9	< 0.001	0.274	Moderate
Previous outcome	3	4,230.5	2	< 0.001	0.320	Moderate

Strength labels follow common rule-of-thumb bands for Cramér's V and are a guide, not a rule. All seven p-values stay below 0.001, so they remain significant even after a conservative Bonferroni correction for seven tests (threshold about 0.007). Because the sample is large, significance alone says little: **effect size** shows that previous outcome, month and channel carry the most association, while weekday, though "significant", is practically negligible.

11.3 Channel gap and contact-frequency trend

- **Channel:** the cellular rate exceeds the telephone rate by **9.51 percentage points** (95% interval 8.95 to 10.06). The odds of subscribing are 3.13 times higher for cellular records, and the relative rate is 2.82.
- **Contact frequency:** a Cochran–Armitage trend test across contact counts 1 to 5+ gives $z = -13.75$ ($p < 0.001$), confirming a downward trend in response as contacts increase.

11.4 Interpretation and caveats

- Chi-square tests assume independent records. Repeat contacts to the same customer and the 12 duplicate rows may violate this, so p-values should be read as indicative rather than exact.
- The tests describe association. Differences may reflect confounding (for example channel, month and prior history moving together) and cannot be read as effects.
- Small groups (Dec, Mar, Sep, Oct, Unknown job) have wide intervals; their rates are useful as leads for investigation, not as stable targets.

12. Project Explanation and Dashboard Working

The Streamlit dashboard supports descriptive exploration. Categorical selections and numeric ranges in the sidebar define a cohort. KPI cards, response summaries, charts, preview rows and findings are recalculated for that selection. Clearing filters returns the complete dataset.

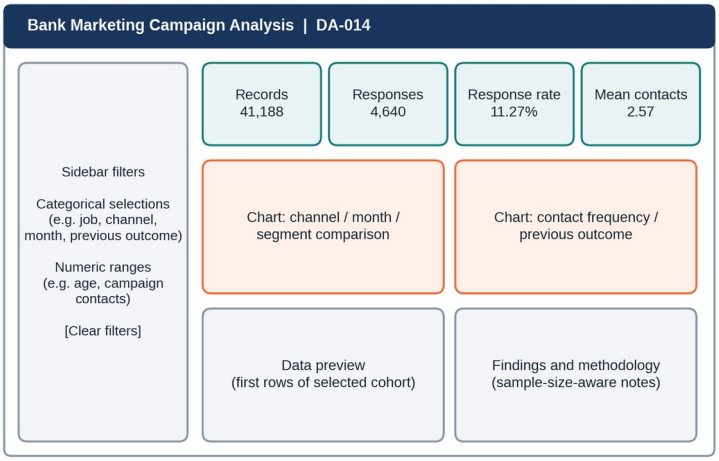


Figure 11. *Dashboard layout (schematic): sidebar filters, KPI cards, charts, data preview and findings.*

Table 18. *Dashboard sections*

Dashboard section	Purpose
Overview	Record count, responses, rate, mean contacts, response comparisons
Filters	Categorical values and ranges over available numeric fields
Charts	Channel, period, segment and frequency comparisons
Data preview	Example rows from the selected data
Findings	Sample-size-aware descriptive summaries
Methodology	Target, denominator, provenance and limitations

12.1 Worked example: a filtered cohort

The saved notebook example for **cellular contacts and ages 30 to 60 inclusive** was independently recalculated: 21,508 records, 2,670 yes responses, a 12.41% response rate (95% interval 11.98% to 12.86%) and 2.44 mean campaign contacts. This is below the cellular-only rate of 14.74%, which illustrates how the headline KPIs change when the cohort changes.

12.2 Demonstration steps

1. Open the dashboard and read the overview cards: 41,188 records, 4,640 responses, 11.27%, 2.57 mean contacts.
2. Select channel = cellular and an age range of 30 to 60; confirm 21,508 records, 2,670 responses, 12.41% and 2.44.
3. Review the channel, month, segment and contact-frequency charts for the filtered cohort.
4. Open the data preview and findings to see sample-size-aware notes.
5. Clear all filters and confirm that 41,188 records and 11.27% return.

13. Python Program and Important Code

The snippets below come from the shared **analysis.py** implementation used by the dashboard. They show the KPI denominator, segment aggregation and contact-frequency grouping. `TARGET_COLUMN` holds the target column name (y) and `CAMPAIGN_BUCKETS` holds the band labels 1, 2, 3, 4 and 5+.

13.1 KPI calculation

```
def calculate_kpis(data: pd.DataFrame) -> dict[str, int | float]:
    """Calculate the headline metrics for the currently selected records.

    Response rate = records with y == 'yes' / records with valid y in the
    current cohort. Rows represent campaign contact records, not unique people.
    """
    records = int(len(data))
    responses = int(data[TARGET_COLUMN].eq("yes").sum()) if records else 0
    rate = responses / records if records else float("nan")
    average_contacts = (
        float(data["campaign"].mean())
        if records and "campaign" in data.columns
        else float("nan")
    )
    return {
        "records": records,
        "responses": responses,
        "response_rate": rate,
        "average_campaign_contacts": average_contacts,
    }
```

13.2 Segment rate calculation

```
def response_by_segment(data: pd.DataFrame, column: str) -> pd.DataFrame:
    """Return record counts, successful responses, and response rates by group."""
    if column not in data.columns:
        raise KeyError(f"Column '{column}' is not in the current dataset.")
    if data.empty:
        return pd.DataFrame(
            columns=["segment", "records", "responses", "response_rate"]
        )
    table = (
        data.groupby(column, dropna=False, observed=True)[TARGET_COLUMN]
        .agg(records="size", responses=lambda values: int(values.eq("yes").sum()))
        .reset_index()
        .rename(columns={column: "segment"})
    )
    table["response_rate"] = table["responses"] / table["records"]
    table["segment"] = table["segment"].astype("string").fillna("Missing")
    return table.sort_values(
        ["response_rate", "records"], ascending=[False, False]
    ).reset_index(drop=True)
```

13.3 Contact-frequency aggregation

```
def contact_frequency_table(data: pd.DataFrame) -> pd.DataFrame:
    """Summarize outcomes by campaign contact-frequency bands.

    The UCI ``campaign`` field is the number of contacts to that record/client
    during this campaign, including the last contact.
    """
    if data.empty or "campaign" not in data.columns:
        return pd.DataFrame(
            columns=["contact_band", "records", "responses", "response_rate"]
        )
    campaign = pd.to_numeric(data["campaign"], errors="coerce")
    bands = pd.Series(
        np.select(
            [campaign.eq(1), campaign.eq(2), campaign.eq(3),
```

```
            campaign.eq(4), campaign.ge(5)],
            CAMPAIGN_BUCKETS,
            default="Missing/other",
        ),
        index=data.index,
        dtype="string",
    )
    temporary = data.assign(_contact_band=bands)
    table = (
        temporary.groupby("_contact_band", observed=True)[TARGET_COLUMN]
        .agg(records="size", responses=lambda values: int(values.eq("yes").sum()))
        .reset_index()
        .rename(columns={"_contact_band": "contact_band"})
    )
    table["response_rate"] = table["responses"] / table["records"]
    order = {
        label: index
        for index, label in enumerate(CAMPAIGN_BUCKETS + ["Missing/other"])
    }
    table["_order"] = table["contact_band"].map(order)
    return (
        table.sort_values("_order")
        .drop(columns="_order")
        .reset_index(drop=True)
    )
```

Invalid target rows are excluded during cleaning. The analysis computes frequencies and rates; it does not fit a machine-learning model.

14. Results and Discussion

Of 41,188 valid-target contact records, 4,640 have y = yes. The full-data response rate is 11.27%. The table below summarises the main results; each one is a historical, unadjusted association.

Table 19. *Summary of key results*

Comparison	Result	Reading
Overall rate	11.27% (4,640 / 41,188)	About 1 in 9 records subscribed
Channel	Cellular 14.74% vs telephone 5.23%	9.5 point gap; V = 0.145
Contact count	13.04% (1 contact) to 6.14% (5+)	Steady decline; trend z = −13.75
Previous outcome	Success 65.11% / failure 14.23% / none 8.83%	Strongest association; V = 0.320
Month	Mar 50.55% … May 6.43%	Large spread but unequal sizes; V = 0.274
Weekday	9.95% (Mon) to 12.12% (Thu)	Statistically detectable, practically negligible
Job	Student 31.43% … blue-collar 6.89%	Wide spread; small to moderate effect
Age band	Under 30 16.26%; 60+ 39.56%	U-shaped; small to moderate effect

14.1 Where effort goes versus where responses come from

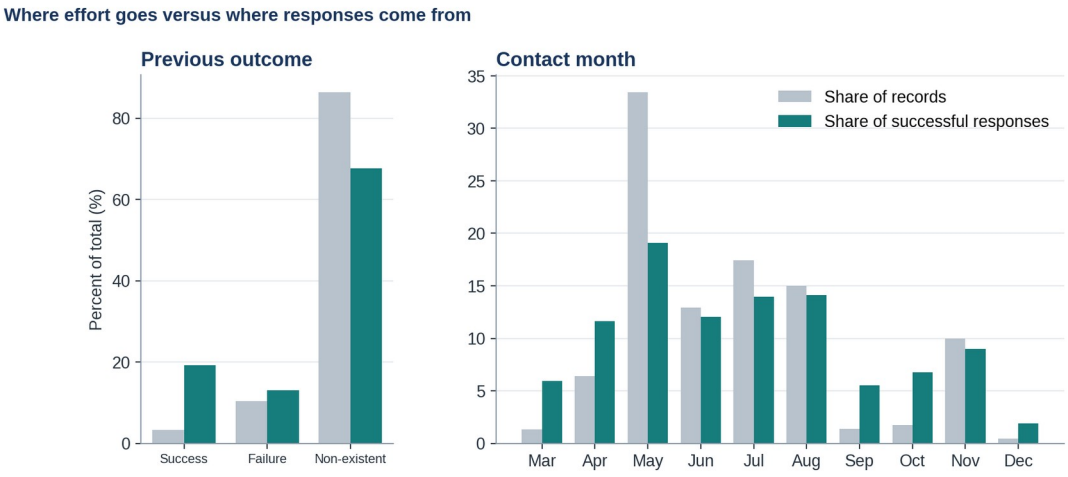


Figure 12. *Share of records versus share of successful responses by previous outcome and by month.*

May carries the largest share of contacts (33.4%) but returns a smaller share of responses (19.1%); previous-success records are rare (3.3%) but produce 19.3% of responses. The gap between the two bars is a simple, interpretable measure of yield per unit of effort. It does not show that moving effort between groups would change outcomes.

14.2 Discussion

Channel and frequency. Cellular records respond at almost three times the telephone rate, and rates decline steadily as the contact count rises. The two patterns may be linked: the data do not show whether telephone-only customers received more attempts, or whether repeated attempts reflect harder-to-reach customers.

Timing. Monthly rates vary from 6.43% to 50.55%, but the highest rates come from months with few records, so the file's volume is concentrated in months with lower yield. The analysis does not establish seasonality; month may stand in for campaign waves, targeting policy or the economic conditions recorded in the dataset.

Customers. Previous success, student or retired status and ages under 30 or over 60 are all linked with higher response rates. These are overlapping descriptors, so they cannot be treated as independent drivers. Using age or occupation to decide who receives outreach also raises fairness considerations that go beyond this descriptive analysis.

14.3 Reconciliation checks

Table 20. *Every grouped table sums back to the full dataset*

Grouping	Σ records	Σ responses	Check
Channel	41,188	4,640	Matches
Contact count	41,188	4,640	Matches
Month	41,188	4,640	Matches
Weekday	41,188	4,640	Matches
Previous outcome	41,188	4,640	Matches
Job	41,188	4,640	Matches
Age band	41,188	4,640	Matches

Direct data calculations match the analysis definitions, and saved notebook outputs were reviewed against them.

15. Findings and Recommendations

15.1 Calculated findings

- Overall rate: 11.27% (4,640 of 41,188 records).
- Cellular and telephone record rates are 14.74% and 5.23%, a gap of 9.51 percentage points.
- Contact-count bands range from 13.04% for one contact to 6.14% for 5+.
- Previous-success records have 65.11%; previous-failure 14.23%; nonexistent history 8.83%.
- Monthly volumes vary substantially (182 to 13,769 records); interpret percentages with counts and intervals.
- Students, retired customers and the 60+ age band have the highest segment rates.

15.2 Recommendations

The observed gaps are hypotheses for further study, not policy directives. The table lists what each observation could suggest and how it could be tested safely.

Table 21. *Observations and suggested follow-up*

Observation	Suggested follow-up (to be tested)
Cellular 14.74% vs telephone 5.23%	Run a controlled pilot comparing channels on comparable customers before shifting volume.
Rate declines after 3 to 4 contacts	Review contact caps and test whether fewer attempts reduce cost without losing responses.
Previous success 65.11%	Prioritise warm leads in a pilot, and check whether the effect holds in recent data.
High volume, low yield in May	Examine why volume was concentrated in May; test spreading outreach across months.
Students, retired, 60+ respond more	Design tailored messages as a test; review fairness and suitability before targeting by age.
Weekday effect negligible	Do not prioritise weekday scheduling on this evidence alone.

15.3 Good practice for presenting these results

- Show counts with rates, and intervals where groups are small; do not rank small groups using percentages alone.
- Demonstrate filters, then clear them and confirm 41,188 records and 11.27% return.
- Before operational use, use recent data, study confounding, and review fairness, privacy and contact safeguards.
- For any future model, define the prediction time and avoid information known only after a call, such as duration.

16. Challenges and Limitations

- **Age of data.** The records cover Portugal from May 2008 to November 2010 and may not represent current campaigns or customers.
- **No customer ID.** Counts are records, not verified distinct people, so repeated contacts to one customer cannot be separated.
- **Duplicates.** 12 exact duplicate rows are retained because the available schema cannot establish whether they are erroneous.
- **Unknown categories.** Unknown values are retained; segment rates are unadjusted and group sizes differ greatly.
- **Statistics.** Intervals and chi-square tests are reported, but they assume independent records and describe association only. No causal design, multivariable adjustment or external validation is implemented.
- **No predictive model.** No machine-learning model, accuracy, precision, recall or predictive score is claimed.
- **Call duration.** Duration is measured after a call and would leak post-contact information in a pre-call prediction task.
- **Verification scope.** Saved notebook outputs and direct calculations were checked; a live hosted-dashboard retest was outside the scope of this report.
- **Dashboard evidence.** The dashboard is illustrated by a layout schematic; screenshots from a live run can be added when the dashboard is hosted.

17. Future Scope

- Refresh the analysis with recent, approved campaign data and clear record definitions and date ranges.
- Study whether segment differences persist after accounting for overlapping attributes, using multivariable methods such as logistic regression, while considering fairness and privacy.
- If prediction is introduced, use a time-aware design, avoid post-contact leakage (such as duration), validate temporally and report calibration and class-specific measures (precision, recall, F1) rather than accuracy alone.
- Incorporate the economic-context fields (employment variation, Euribor, confidence indices) to examine how conditions relate to response.
- Add exports of filtered tables, accessibility improvements, data-quality checks and a deployment health check to the dashboard.
- Prepare an offline demonstration fallback and document hosting requirements.

18. Conclusion

The DA-014 prototype turns a public bank marketing dataset into an inspectable descriptive dashboard and notebook analysis. It defines response rate consistently as yes records divided by valid-target records and reports 4,640 ÷ 41,188 = 11.27%. Channel, month, previous-outcome, segment and contact-frequency comparisons make historical patterns easier to explore, and Wilson intervals, chi-square tests and effect sizes show which differences are large, which are small and which rest on small samples.

The main contribution is a filterable summary with transparent counts, quantified uncertainty and explicit limitations. Findings are descriptive, not causal, and no predictive model was implemented. Further work should use current data and evaluate confounding, fairness and customer safeguards before any operational use.

19. References

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Appendix A: Dataset Field Dictionary

Fields of bank-additional-full.csv as documented by the UCI repository [3].

Table 22. *Field dictionary*

Field	Description
age	Age of the customer (numeric)
job	Type of job (categorical, includes unknown)
marital	Marital status (divorced, married, single, unknown)
education	Education level (categorical, includes unknown)
default	Has credit in default? (yes / no / unknown)
housing	Has a housing loan? (yes / no / unknown)
loan	Has a personal loan? (yes / no / unknown)
contact	Contact communication type (cellular / telephone)
month	Last contact month of year
day_of_week	Last contact day of the week
duration	Last contact duration in seconds; known only after the call, so unsuitable for pre-call prediction
campaign	Number of contacts during this campaign for this client, including the last contact
pdays	Days since the client was last contacted in a previous campaign (999 = not previously contacted)
previous	Number of contacts before this campaign for this client
poutcome	Outcome of the previous campaign (failure / nonexistent / success)
emp.var.rate	Employment variation rate (quarterly indicator)
cons.price.idx	Consumer price index (monthly indicator)
cons.conf.idx	Consumer confidence index (monthly indicator)
euribor3m	Euribor 3-month rate (daily indicator)
nr.employed	Number of employees (quarterly indicator)
y	Target: has the client subscribed a term deposit? (yes / no)

Appendix B: Reproducibility Notes

The project artefacts used for this report are listed below.

Table 23. *Project files*

File	Role
bank-additional-full.csv	Source data (UCI), 41,188 records, semicolon-delimited
analysis.py	Shared analysis module: cleaning, KPIs, filters, group rates
app.py	Streamlit dashboard
bank_marketing_analysis.ipynb	Notebook analysis using the shared module
README	Project description and usage notes
Chart images	Static figures for the core comparisons

B.1 Script for the statistical checks

Report-stage script used for the Wilson intervals and chi-square tests in Section 11. It is applied to the grouped counts (records, yes) and is not part of the dashboard.

```
from scipy import stats
import numpy as np

def wilson_interval(successes, n, z=1.959964):
    p = successes / n
    d = 1 + z * z / n
    centre = (p + z * z / (2 * n)) / d
    half = z * np.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / d
    return centre - half, centre + half

def chi_square_by_group(groups):          # groups = [(records, yes), ...]
    observed = np.array([[yes, n - yes] for n, yes in groups], float)
    chi2, p, dof, _ = stats.chi2_contingency(observed, correction=False)
    cramers_v = np.sqrt(chi2 / (observed.sum() * (min(observed.shape) - 1)))
    return chi2, dof, p, cramers_v
```

The dashboard is a Streamlit application; the statistical checks can be reproduced from the grouped counts using the script above.